Probability and Inference Cheat Sheet

Material originally compiled by William Chen and Joe Blitzstein. Substantially expanded with excerpts from the Statistics Cookbook by Matthias Vallentin, translated and adapted maintaining mathematical rigor. By Roberto León Landeros. This work is distributed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC BY-NC-SA 4.0).

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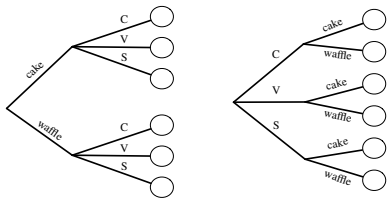
Counting and Combinatorics

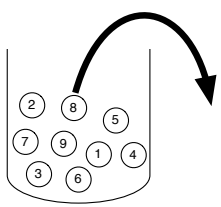
Multiplication Rule



For a compound experiment with multiple components, if the i-th component has n_i possible outcomes, the total number of outcomes for the full experiment of r components is the product of the individual cardinalities:

$$N = n_1 \times n_2 \times \cdots \times n_r = \prod_{i=1}^{r} n_i$$

Sampling Table



Options for drawing a sample of size n from a population $S = \{1, \ldots, s\}$ of size s:

Sample n from pop. s	With Replacement	Without Replacement
Ordered	s^n	$(s)_n = \frac{s!}{(s-n)!}$
Unordered	$\binom{n+s-1}{s-1}$	$\binom{s}{n} = \frac{s!}{n!(s-n)!}$

Partitions and Allocations (Balls in Bins)

Stirling Numbers of the Second Kind $\left\{ {n \atop k} \right\}$: Ways to partition n elements into k non-empty subsets. $\left\{ {n \atop k} \right\} = k \left\{ {n-1 \atop k} \right\} + \left\{ {n-1 \atop k-1} \right\}$.

Allocation of n balls into m bins:

Balls	Bins	Any allocation	Sampling w/o repl. (Inj.)
Distinct	Distinct	m^n	$\frac{m!}{(m-n)!}$
Indist.	Distinct	$\binom{m+n-1}{n}$	$\binom{m}{n}$
Distinct	Indist.	$\sum_{k=1}^{m} \left\{ {n \atop k} \right\}$	1 if $m \geq n$

Additional Definitions of Probability

Classical (Laplace) If all elements are equiprobable: $P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}} = \frac{\#A}{\#\Omega}$

Relative Frequency $h_n(A) = \frac{m}{n}$ (absolute frequency over total).

Geometric $P(A) = \frac{m(A)}{m(\Omega)}$, where m denotes the geometric measure (length, area, or volume).

Algebraic Expansions

Binomial Theorem: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$

Multinomial Coefficient: $\binom{n}{n_1, n_2, \ldots, n_k} = \frac{n!}{n_1! n_2! \ldots n_k!}$

Multinomial Theorem: $(a_1 + \cdots + a_m)^n = \sum_{n_1 + \cdots + n_m = n} \binom{n}{n_1, \ldots, n_m} a_1^{n_1} \ldots a_m^{n_m}$

Theoretical Foundations

Set Operations and Algebra

De Morgan's Laws (Duality) $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$

Set Difference $A \setminus B = A \cap B^c$

Symmetric Difference $A \triangle B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$

Union Identities $A \cup B = A \cup (B \cap A^c) = (A \cap B^c) \cup (B \cap A^c) \cup (A \cap B)$

Distributive Properties $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Kolmogorov's Axiomatic System

Let P be a function over a σ-algebra $\mathfrak{F}$:

1. $\Omega \in \mathfrak{F}$
2. $A \in \mathfrak{F} \implies A^c \in \mathfrak{F}$
3. $\bigcup_{i=1}^{\infty} A_i \in \mathfrak{F}$

The axioms of the probability measure P are:

Axiom I (Non-negativity) $0 \leq P(A) \leq 1$

Axiom II (Certainty) $P(\Omega) = 1$

Axiom III (Finite additivity) If $A \cap B = \emptyset$, $P(A \cup B) = P(A) + P(B)$

Axiom IV (σ-additivity) For disjoint $\{A_i\}_{i=1}^{\infty}$, $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

Basic Rules and Theorems

Impossibility $P(\emptyset) = 0$

Complement $P(A^c) = 1 - P(A)$

Monotonicity If $A \subset B \implies P(A) \leq P(B)$

Difference If $A \subset B \implies P(B \setminus A) = P(B) - P(A)$

General Subtraction $P(A \setminus B) = P(A) - P(A \cap B)$

Boole's Inequality $P(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} P(A_i)$

Bonferroni's Ineq. $P(\bigcup_{i=1}^{n} A_i) \geq \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j)$

Continuity If $A_n \uparrow A \implies P(A) = \lim_{n \to \infty} P(A_n)$

Conditional Probability

Independence

Independent Events A and B are independent if knowing that A occurred provides no information about B. Equivalencies:

$$P(A \cap B) = P(A)P(B)$$
$$P(A|B) = P(A)$$
$$P(B|A) = P(B)$$

Conditional Independence A and B are conditionally independent given C if $P(A \cap B|C) = P(A|C)P(B|C)$. This neither implies absolute independence, nor vice versa.

Joint, Marginal, and Conditional Probabilities

Joint Probability $P(A \cap B)$ or $P(A, B)$ – Probability that both A and B occur.

Marginal Probability $P(A)$ – Unconditional probability.

Conditional Probability $P(A|B) = \frac{P(A \cap B)}{P(B)}$ (for $P(B) > 0$). It satisfies all of Kolmogorov's axioms.

Probability of Intersections and Unions

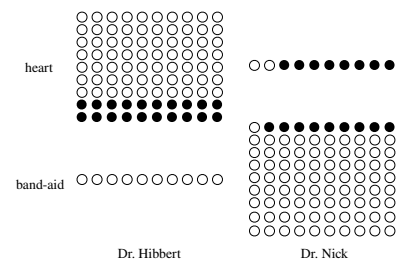
Chain Rule (Multiplication)

$$P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1) \ldots P(A_n|A_1 \ldots A_{n-1})$$

$$P\Big(\bigcap_{i=1}^{n} A_i\Big) = \prod_{i=1}^{n} P\Big(A_i \Big| \bigcap_{j=1}^{i-1} A_j\Big)$$

Inclusion-Exclusion (Unions) For 2 events:
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
For n events: $P(\bigcup_{i=1}^{n} A_i) = \sum_{r=1}^{n} (-1)^{r-1} S_r^n$
Where $S_r^n = \sum_{1 \leq i_1 < \cdots < i_r \leq n} P(A_{i_1} \cap \cdots \cap A_{i_r})$

Simpson's Paradox



It is analytically possible to observe:

$$P(A|B, C) < P(A|B^c, C) \text{ and } P(A|B, C^c) < P(A|B^c, C^c)$$
$$\text{but globally: } P(A|B) > P(A|B^c).$$

Law of Total Probability (LOTP)

Let $\{B_1, \ldots, B_n\}$ be a partition of the sample space (mutually disjoint, union equal to Ω).

$$P(A) = P(A|B_1)P(B_1) + \cdots + P(A|B_n)P(B_n) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$

$$P(A) = P(A \cap B_1) + \cdots + P(A \cap B_n) = \sum_{i=1}^{n} P(A \cap B_i)$$

For **LOTP with extra conditioning on** C:

$$P(A|C) = \sum_{i=1}^{n} P(A|B_i, C)P(B_i|C)$$

Bayes' Theorem

Conditional Inversion:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

With additional conditioning:

$$P(A|B,C) = \frac{P(B|A,C)P(A|C)}{P(B|C)}$$

Odds Form of Bayes' Rule:

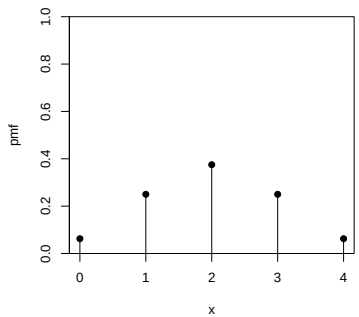
$$\frac{P(A|B)}{P(A^c|B)} = \frac{P(B|A)}{P(B|A^c)}\frac{P(A)}{P(A^c)}$$

Discrete Random Variables

PMF, CDF, and Independence

Probability Mass Function (PMF) Yields the probability that a *discrete* R.V. takes a specific value x.

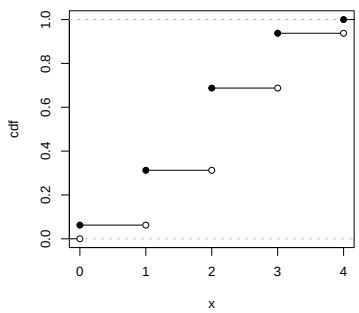
$$p_X(x) = P(X = x)$$



Conditions: $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$.

Cumulative Distribution Function (CDF) Probability that an R.V. is less than or equal to x.

$$F_X(x) = P(X \leq x)$$



Monotonically increasing, right-continuous, $\lim_{x\to-\infty} F_X(x) = 0$, $\lim_{x\to\infty} F_X(x) = 1$.

Independence of R.V.s R.V.s X and Y are independent if knowing the value of one provides no information about the other. For discrete variables, for all x, y: $P(X = x, Y = y) = P(X = x)P(Y = y)$

Expected Value and Indicators

Expectation and Linearity

Expectation Central measure or probability-weighted average of the distribution. For discrete X with support $\{x_1, x_2, \dots\}$:

$$E(X) = \sum_i x_i P(X = x_i)$$

X	Y	$X+Y$
3	4	7
2	2	4
6	8	14
10	23	33
1	–3	–2
1	0	1
5	9	14
4	1	5
...	...	...

$$\frac{1}{n}\sum_{i=1}^{n} x_i \quad + \quad \frac{1}{n}\sum_{i=1}^{n} y_i \quad = \quad \frac{1}{n}\sum_{i=1}^{n} (x_i + y_i)$$

$$E(X) \quad + \quad E(Y) \quad = \quad E(X+Y)$$

Linearity For R.V.s X, Y and constants a, b, c:

$$E(aX + bY + c) = aE(X) + bE(Y) + c$$

Identically distributed If X and Y share the same distribution, $E(X) = E(Y)$ and $E(g(X)) = E(g(Y))$.

Conditional Expected Value Weighted expectation over a conditional event A:

$$E(X|A) = \sum_x xP(X = x|A)$$

Indicator Variables

Takes the value 1 if the event occurs, 0 otherwise.

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{if } A^c \text{ occurs} \end{cases}$$

Properties: $I_A^2 = I_A$, $I_A I_B = I_{A\cap B}$, $I_{A\cup B} = I_A + I_B - I_A I_B$.
Distribution: $I_A \sim \text{Bern}(p)$ with $p = P(A)$.
Fundamental Bridge: $E(I_A) = P(A)$.
Variance and Covariance: $\text{Var}(I_A) = P(A)(1 - P(A))$, $\text{Cov}(I_A, I_B) = P(A \cap B) - P(A)P(B)$.

Variance and Standard Deviation

$$\text{Var}(X) = E\left((X - E(X))^2\right) = E(X^2) - (E(X))^2$$

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Continuous R.V.s, LOTUS, UoU

Continuous Random Variables (CRVs)

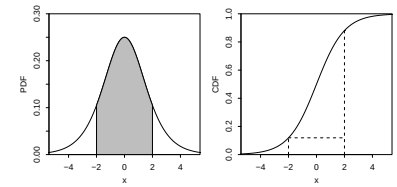
Probability over intervals

$$P(a \leq X \leq b) = F_X(b) - F_X(a)$$

For the Standard Normal $Z \sim \mathcal{N}(0,1)$, the probability is $\Phi(b) - \Phi(a)$.

Probability Density Function (PDF) The derivative of the CDF.

$$f(x) = \frac{d}{dx}F(x) \implies F(x) = \int_{-\infty}^{x} f(t)dt$$



Must integrate to 1 and be non-negative.

Expected Value Analogous to the discrete case but with an integral.

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

LOTUS (Law of the Unconscious Statistician)

Facilitates finding the expected value of a function dependent on X, $g(X)$, by operating over the PDF or PMF of X. Discrete: $E(g(X)) = \sum_x g(x)P(X = x)$ Continuous: $E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$

Universality of the Uniform (UoU)

Evaluating a continuous R.V. in its own CDF results in a Uniform distribution. If X has CDF F, then $F(X) \sim \text{Unif}(0,1)$. Conversely, if $U \sim \text{Unif}(0,1)$, then $F^{-1}(U)$ has distribution F. This facilitates computational simulation of R.V.s.

Moments and Generating Functions

Moments

Describe the shape of the distribution. For standardized R.V. $Z = \frac{X-\mu}{\sigma}$:

Mean (1st moment) $E(X) = \mu_1$

Variance (2nd central) $\text{Var}(X) = \mu_2 - \mu_1^2$

Skewness $\text{Skew}(X) = E(Z^3)$

Kurtosis $\text{Kurt}(X) = E(Z^4) - 3$

Moment Generating Function (MGF)

MGF Defined as $M_X(t) = E(e^{tX})$.

Obtaining Moments The k-th derivative evaluated at 0 yields the k-th moment: $\mu_k = E(X^k) = M_X^{(k)}(0)$. Based on Taylor expansion: $M_X(t) = \sum_{k=0}^{\infty} \frac{E(X^k)t^k}{k!}$.

Linear Transformation For $Y = aX + b$, $M_Y(t) = e^{bt}M_X(at)$.

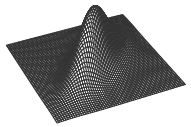
Uniqueness The MGF uniquely determines the probability distribution (if it exists).

Sum of Independent R.V.s If $X \perp\!\!\!\perp Y$: $M_{X+Y}(t) = M_X(t)M_Y(t)$

Joint Density and Mass Functions

Joint Distributions

Joint CDF: $F_{X,Y}(x,y) = P(X \le x, Y \le y)$
Joint PMF: $p_{X,Y}(x,y) = P(X = x, Y = y)$
Joint PDF: $f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$



Conditional Distributions and Bayes' Rule

For discrete: $P(Y = y|X = x) = \frac{P(X=x,Y=y)}{P(X=x)} = \frac{P(X=x|Y=y)P(Y=y)}{P(X=x)}$

For continuous: $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{f_{X|Y}(x|y)f_Y(y)}{f_X(x)}$

Marginal Distributions

Obtaining an isolated distribution by integrating/summing out the unwanted variable. Discrete: $P(X = x) = \sum_y P(X = x, Y = y)$
Continuous: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy$

Multivariate LOTUS

Discrete: $E(g(X,Y)) = \sum_x \sum_y g(x,y)P(X = x, Y = y)$
Continuous: $E(g(X,Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y)f_{X,Y}(x,y)dxdy$

Covariance and Transformations

Covariance and Correlation

Covariance Multivariate analog of variance.

$$\mathrm{Cov}(X,Y) = E((X - E(X))(Y - E(Y))) = E(XY) - E(X)E(Y)$$

Correlation Standardized covariance $[-1, 1]$.

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

Independence $X \perp\!\!\!\perp Y \implies \mathrm{Cov}(X,Y) = 0$. The converse is generally false (with exceptions like the Multivariate Normal).

Variance of Sums $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$
For a generic sum with identically related variables (symmetry):

$$\mathrm{Var}\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$$

Bilinear Properties $\mathrm{Cov}(aX + c, bY + d) = ab\mathrm{Cov}(X,Y)$
$\mathrm{Cov}(X+Y, Z+W) = \mathrm{Cov}(X,Z) + \mathrm{Cov}(X,W) + \mathrm{Cov}(Y,Z) + \mathrm{Cov}(Y,W)$

1D Transformation

For $Y = g(X)$ where g is monotonic and differentiable:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Multivariate Transformations

For R.V. (U,V) with a joint PDF and $(X,Y) = g(U,V)$, the determinant of the Jacobian matrix is fundamental to change the density domain.

$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}$$

$$f_{X,Y}(x,y) = f_{U,V}(u,v)|\det(J)|$$

Convolutions

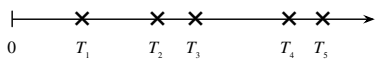
Integral for the PDF of the sum of two independent continuous variables X, Y:

$$f_{X+Y}(t) = \int_{-\infty}^{\infty} f_X(x)f_Y(t-x)dx$$

Poisson Process

Definition Event rates modeled in temporal intervals λ.

1. Events in an interval t follow $\mathrm{Pois}(\lambda t)$.
2. Increments in disjoint intervals are independent.



Superposition and Thinning

1. **Superposition:** If $N_1(t) \sim PP(\lambda_1)$ and $N_2(t) \sim PP(\lambda_2)$ are independent, then $N_1(t) + N_2(t) \sim PP(\lambda_1 + \lambda_2)$.
2. **Thinning:** If $N \sim \mathrm{Pois}(\lambda)$ and each event is classified as type i with prob p_i, then the counts N_i are independent and $N_i \sim \mathrm{Pois}(\lambda p_i)$.

Count-Time Duality The time until the first event follows an Exponential distribution $\mathrm{Expo}(\lambda)$, since the event of waiting t hours is equivalent to observing 0 arrivals in t hours ($N_t = 0$). Inter-arrival times are i.i.d. $\mathrm{Expo}(\lambda)$.

Order Statistics

Definition Let $X_1, \ldots, X_n$ be i.i.d. random variables. Sorting them from smallest to largest, $X_{(i)}$ denotes the i-th order statistic. The minimum is $X_{(1)}$ and the maximum $X_{(n)}$. They usually exhibit statistical dependence.

Distribution The CDF and PDF of element $X_{(i)}$ are given by:

$$F_{X_{(i)}}(x) = \sum_{k=i}^{n} \binom{n}{k} F(x)^k (1 - F(x))^{n-k}$$

$$f_{X_{(i)}}(x) = n\binom{n-1}{i-1} F(x)^{i-1}(1 - F(x))^{n-i} f(x)$$

Uniform Order Statistics The j-th component of $U_1, \ldots, U_n \sim \mathrm{Unif}(0,1)$ distributes as $U_{(j)} \sim \mathrm{Beta}(j, n-j+1)$.

Conditional Expectation

Conditioned on an Event Expectation given event A. Results in a deterministic scalar value.

Discrete	Continuous
$E(Y) = \sum_y yP(Y = y)$	$E(Y) = \int_{-\infty}^{\infty} yf_Y(y)dy$
$E(Y\|A) = \sum_y yP(Y = y\|A)$	$E(Y\|A) = \int_{-\infty}^{\infty} yf(y\|A)dy$

Conditioned on an R.V. The term $E(Y|X)$ is a mathematical function of the random variable X. It only collapses to a constant if they are independent ($X \perp\!\!\!\perp Y$).

Analytical Properties

1. $E(Y|X) = E(Y)$ if $X \perp\!\!\!\perp Y$.
2. $E(h(X)W|X) = h(X)E(W|X)$ (pulling out known scalars).
3. $E(E(Y|X)) = E(Y)$ (Adam's Law).

Adam's Law (Law of Total Expectation)

$$E(Y) = \sum_{i=1}^{n} E(Y|A_i)P(A_i)$$

Eve's Law (Law of Total Variance)

$$\mathrm{Var}(Y) = E(\mathrm{Var}(Y|X)) + \mathrm{Var}(E(Y|X))$$

Classical Limit Theorems

Types of Convergence

In Distribution (Law) $X_n \xrightarrow{D} X \iff \lim_{n\to\infty} F_n(t) = F(t)$

In Probability $X_n \xrightarrow{P} X \iff \lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0$

Almost Surely (a.s.) $X_n \xrightarrow{a.s.} X \iff P(\lim_{n\to\infty} X_n = X) = 1$

In Mean Square (L_2) $X_n \xrightarrow{qm} X \iff \lim_{n\to\infty} E((X_n - X)^2) = 0$

Hierarchy: $qm \implies P \implies D$ and $a.s. \implies P$.

Slutsky's Theorem

If $X_n \xrightarrow{D} X$ and $Y_n \xrightarrow{P} c$:

$$X_n + Y_n \xrightarrow{D} X + c \quad \text{and} \quad X_nY_n \xrightarrow{D} cX$$

Law of Large Numbers (LLN)

For a large set of i.i.d. R.V.s X_i with mean μ, the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ converges directly to μ as $n \to \infty$ (in probability for WLLN and almost surely for SLLN).

Central Limit Theorem (CLT)

Sums and sample averages of R.V.s converge asymptotically to the Normal distribution, regardless of their original distribution.

$$Y = \sum X_i \dot{\sim} \mathcal{N}(n\mu_X, n\sigma_X^2)$$

$$\bar{X}_n \dot{\sim} \mathcal{N}(\mu_X, \sigma_X^2/n)$$

Asymptotic Standardization:

$$\frac{\sqrt{n}(\bar{X}_n - \mu_X)}{\sigma_X} \xrightarrow{D} \mathcal{N}(0,1)$$

Statistical Inference

Point Estimation and Properties

Bias $\mathrm{Bias}(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta$.

Consistency $\hat{\theta}_n \xrightarrow{P} \theta$.

Mean Squared Error (MSE) $E((\hat{\theta}_n - \theta)^2) = \mathrm{Bias}^2 + \mathrm{Var}(\hat{\theta}_n)$.

Sufficiency $T(X)$ is sufficient if the conditional distribution of the data given T does not depend on θ.

Factorization Theorem (Fisher-Neyman) $T(X)$ is sufficient for $\theta \iff f(x|\theta) = g(T(x), \theta)h(x)$.

Asymptotic Efficiency An estimator is efficient if its variance achieves the Cramér-Rao Lower Bound.

Rao-Blackwell Improvement Let $\hat{\theta}$ be an unbiased estimator and T a sufficient statistic. Then $\eta = E(\hat{\theta}|T)$ is an unbiased estimator with $\mathrm{Var}(\eta) \le \mathrm{Var}(\hat{\theta})$.

Fisher Information and Lower Bound

Fisher Information $I(\theta) = -E\left(\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\right)$.

Cramér-Rao Lower Bound For any unbiased estimator $W(X)$ of θ:

$$\mathrm{Var}(W) \geq \frac{1}{nI(\theta)}$$

Confidence Intervals (CI)

Mean (known var) $\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

Mean (unknown var) $\bar{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$

Proportion $\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Method of Moments (MoM)

Equate theoretical population moments $\alpha_j(\theta) = E(X^j)$ with sample moments $\hat{\alpha}_j = \frac{1}{n}\sum X_i^j$ and solve for θ.

Maximum Likelihood Estimation (MLE)

Likelihood Function $L(\theta) = \prod_{i=1}^n f(X_i;\theta)$

Log-Likelihood $\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(X_i;\theta)$

MLE Estimator $\hat{\theta}_n = \sup_\theta L(\theta)$.

Asymptotic Properties Under regularity conditions: $\hat{\theta}_{MLE} \xrightarrow{D} \mathcal{N}(\theta, \frac{1}{nI(\theta)})$. This connects MLE with the Cramér-Rao Bound and explains why it is "efficient".

Fisher Information
$I(\theta) = \mathrm{Var}\left(\frac{\partial}{\partial\theta}\log f(X;\theta)\right) = -E\left(\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\right)$. For the entire sample: $I_n(\theta) = nI(\theta)$.

Delta Method

If $\tau = g(\theta)$ where g is differentiable and $g'(\theta) \neq 0$:

$$\frac{g(\hat{\theta}_n) - g(\theta)}{|g'(\hat{\theta}_n)|\hat{\mathrm{se}}(\hat{\theta}_n)} \xrightarrow{D} \mathcal{N}(0,1)$$

Empirical Distribution and Bootstrap

Empirical CDF (ECDF) $\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^n I(X_i \leq x)$.

DKW Inequality Non-parametric confidence bound for F: $P(\sup_x |F(x) - \hat{F}_n(x)| > \varepsilon) \leq 2e^{-2n\varepsilon^2}$.

Bootstrap Resampling with replacement to estimate the variance of a statistic $T_n = g(X_1,\ldots,X_n)$.

Hypothesis Testing

Level and Power Type I Error (α): Reject H_0 when true. Type II Error (β): Fail to reject H_0 when false. Power: $1-\beta$.

p-value Probability of obtaining a test statistic at least as extreme as the one observed, assuming H_0 is true.

Neyman-Pearson Lemma: For simple H_0 vs H_1, the test with critical region $\{\frac{L(\theta_1)}{L(\theta_0)} > k\}$ is the most powerful.

Classical Tests

Wald Test $W = \frac{\hat{\theta}-\theta_0}{\hat{\mathrm{se}}}$. Reject H_0 if $|W| > z_{\alpha/2}$.

Likelihood Ratio Test (LRT) Compares the model under H_0 against the full model. Statistic $\lambda = 2\log\left(\frac{L(\hat{\theta})}{L(\hat{\theta}_0)}\right)$.

Multinomial LRT For goodness of fit:
$\lambda(X) = 2\sum_{j=1}^k X_j \log\left(\frac{\hat{p}_j}{p_{0j}}\right) \xrightarrow{D} \chi^2_{k-1}$.

Wilks' Theorem: Under H_0 and regularity conditions, $\lambda \xrightarrow{D} \chi^2_d$ where d is the difference in free parameters.

Pearson's Chi-Squared Test $T = \sum_{j=1}^k \frac{(X_j - E(X_j))^2}{E(X_j)} \xrightarrow{D} \chi^2_{k-1}$. For independence in $I \times J$ contingency tables, $df = (I-1)(J-1)$.

Student's t-Test $T = \frac{\bar{X}-\mu_0}{s/\sqrt{n}} \xrightarrow{D} t_{n-1}$. For two samples, standard error and degrees of freedom are adapted.

Bayesian Inference

Bayesian Updating

The posterior distribution combines prior information with empirical evidence from the likelihood:

$$f(\theta|x^n) = \frac{f(x^n|\theta)f(\theta)}{\int f(x^n|\theta)f(\theta)d\theta} \propto L(\theta)f(\theta)$$

Likelihood	Parameter	Conjugate Prior
Bernoulli / Binomial	p	Beta(α,β)
Poisson(λ)	λ	Gamma(α,β)
Normal (known σ^2)	μ	Normal(μ_0,σ_0^2)
Normal (known μ)	σ^2	Inv-Gamma
Multinomial$(\vec{p})$	$\vec{p}$	Dirichlet$(\vec{\alpha})$
Exponential(λ)	λ	Gamma(α,β)

Credible Intervals and Bayes Factor

Credible Interval Region R_n such that $P(\theta \in R_n|x^n) = 1-\alpha$. HPD is the Highest Posterior Density region.

Bayes Factor (BF) Ratio of marginal likelihoods of two hypotheses. $\frac{P(H_i|x^n)}{P(H_j|x^n)} = BF_{ij} \times \frac{P(H_i)}{P(H_j)}$.

Regression and Regularization

Simple and Multiple Linear Regression

Model: $Y = X\beta + \epsilon$ with $E(\epsilon) = 0, \mathrm{Var}(\epsilon) = \sigma^2 I$.

Ordinary Least Squares (OLS) $\hat{\beta} = (X^TX)^{-1}X^TY$.

Gauss-Markov Theorem: In a linear model with mean zero, uncorrelated, and homoscedastic errors, the OLS estimator is the BLUE. **Assumptions (LINE):** Linearity, Independence, Normality of errors, Equal variance (Homoscedasticity).

Diagnostics:
Non-Normality $\implies$ Box-Cox Transformation.
Heteroscedasticity $\implies$ Weighted Least Squares (WLS).
Multicollinearity $\implies$ Ridge Regression.

Estimation of σ^2 $\hat{\sigma}^2 = \frac{\mathrm{RSS}}{n-k} = \frac{1}{n-k}\sum(Y_i - \hat{Y}_i)^2$.

Variance of Estimator $\mathrm{Var}(\hat{\beta}|X) = \sigma^2(X^TX)^{-1}$.

R-squared $R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\sum(Y_i-\hat{Y}_i)^2}{\sum(Y_i-\bar{Y})^2}$.

Regularization and Bias-Variance Trade-off

Avoids overfitting by penalizing coefficients β:

Ridge (L_2): $\min \|Y - X\beta\|^2 + \lambda\|\beta\|_2^2$. Reduces variance in presence of multicollinearity.

Lasso (L_1): $\min \|Y - X\beta\|^2 + \lambda\|\beta\|_1$. Performs automatic variable selection (shrinks $\beta_i \to 0$).

MSE Decomposition: $MSE(\hat{\theta}) = \mathrm{Bias}^2(\hat{\theta}) + \mathrm{Var}(\hat{\theta}) + \sigma^2_{irr}$.

Logistic Regression (GLM)

For binary classification problems, predicts probabilities using the logit link function:

$$p(X) = \frac{e^{\beta_0+\beta X}}{1+e^{\beta_0+\beta X}} \implies \mathrm{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta X$$

Model Selection

Penalizes complexity (parameters k) to prevent overfitting.

AIC $AIC = 2k - 2\ell(\hat{\beta},\hat{\sigma}^2)$

BIC $BIC = k\log n - 2\ell(\hat{\beta},\hat{\sigma}^2)$

Mallow's C_p $\hat{R}(S) = \mathrm{RSS} + 2k\hat{\sigma}^2$

Time Series

White Noise $w_t \sim wn(0,\sigma_w^2)$ has $E(w_t) = 0, \mathrm{Var}(w_t) = \sigma_w^2$ and $\gamma(s,t) = 0$ for $s \neq t$.

Strict Stationarity The joint distribution is invariant under time shifts.

Weak Stationarity Constant mean and autocovariance $\gamma(s,t)$ that only depends on the lag $h = |s-t|$.

ACF and PACF Autocorrelation Function $\rho(h) = \frac{\gamma(h)}{\gamma(0)}$. PACF removes the effect of intermediate lags.

Backshift Operator $B^kX_t = X_{t-k}$.

Difference Operator $\nabla^d = (1-B)^d$.

ARIMA (p, d, q) Models

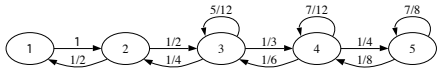
AR(p) Autoregressive $x_t = \sum_{i=1}^p \phi_i x_{t-i} + w_t$. (ACF decays asymptotically, PACF cuts off at p).

MA(q) Moving Average $x_t = w_t + \sum_{j=1}^q \theta_j w_{t-j}$. (ACF cuts off at q, PACF decays).

Integration (d) Differencing $\nabla^d = (1-B)^d$ to remove stochastic trends.

Markov Chains

Definition and Properties



A Markov chain (over a discrete state space) guarantees the fundamental **memoryless** property: given the present state, future and past states are conditionally independent.

$$P(X_{n+1} = j|X_0\ldots,X_n = i) = P(X_{n+1} = j|X_n = i)$$

Recurrent states guarantee their future return, while transient states possess a strict probability of infinite abandonment. States can be periodic or aperiodic according to the GCD of their return steps.

Transition Matrix and Stationary Distribution

Matrix Q where $q_{ij} = P(X_{n+1} = j | X_n = i)$. The transition probability in m steps is the respective element in Q^m.
A PMF distribution $\vec{s}$ is **Stationary** if $\vec{s}Q = \vec{s}$. For irreducible and aperiodic systems, a unique stationary distribution exists. The expected return time to state i is $1/s_i$. If $s_i q_{ij} = s_j q_{ji}$ (Detailed balance equation) for all i, j, the system has Reversibility, which inherently implies stationarity.
Ergodicity: A chain is ergodic if it is irreducible, aperiodic, and all its states are positive recurrent. This guarantees convergence to a unique stationary distribution.

Sampling Methods

Inverse Transform If $U \sim \text{Unif}(0,1)$, then $X = F^{-1}(U)$ has CDF F.

Rejection Sampling To simulate $f(x)$ using $g(x)$ such that $f(x) \leq Mg(x)$. 1. Generate $X \sim g$ and $U \sim \text{Unif}(0,1)$. 2. Accept X if $U \leq \frac{f(X)}{Mg(X)}$.

Importance Sampling Estimates $E_f[h(X)]$ using g: $E_f[h(X)] = \int h(x) \frac{f(x)}{g(x)} g(x) dx = E_g[h(X)w(X)]$ where $w(x) = f(x)/g(x)$.

Decision Theory

Loss Function $L(\theta, a)$ measures the cost of choosing action a when the parameter is θ. Ex: $L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$.

Statistical Risk $R(\theta, \delta) = E_\theta[L(\theta, \delta(X))]$.

Bayes Risk $r(\pi, \delta) = \int R(\theta, \delta)\pi(\theta) d\theta$. The Bayes rule minimizes the average risk.

Minimax Rule Minimizes the maximum possible risk: $\inf_\delta \sup_\theta R(\theta, \delta)$.

Non-parametric Statistics

Kernel Density Estimation (KDE) $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$.

Nadaraya-Watson Regression Estimator of $m(x) = E(Y|X = x)$: $\hat{m}_h(x) = \frac{\sum K_h(x - X_i) Y_i}{\sum K_h(x - X_i)}$.

Continuous Distributions

Uniform Unif(a, b)

Probability purely proportional to the geometric size of the support region, with a flat density and zero outside.

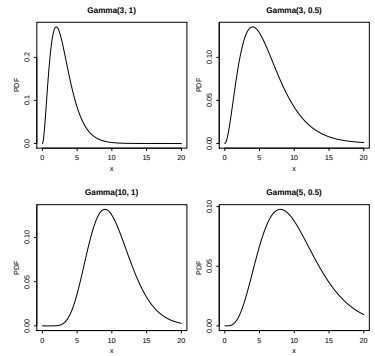
Gaussian Normal $\mathcal{N}(\mu, \sigma^2)$

Symmetric distribution most present in nature due to the Central Limit Theorem. It is invariant under scale transformations and translations (resulting in new Gaussians). The Standard Normal evaluates with $Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$.

Exponential Expo(λ)

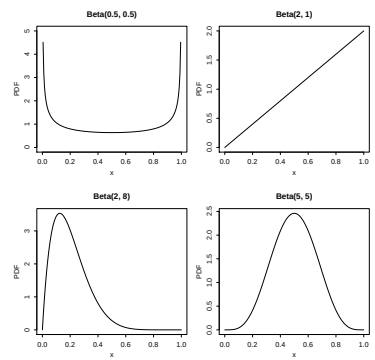
The only continuous distribution with the **memoryless** (Markovian) property. Expresses the waiting time until the next event in Poisson processes; an expected time dictated by $1/\lambda$.
Analytical formulation: $P(X > s + t \mid X > s) = P(X > t)$.

Gamma Gamma(a, λ)



Represents the accumulated waiting time for a independent occurrences in an Exponential λ scheme.

Beta Beta(a, b)



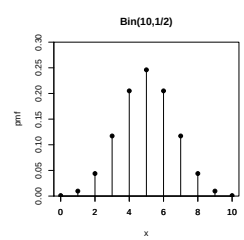
Its domain $(0, 1)$ allows modeling random probabilities and proportions. It is the classic conjugate prior of the Binomial in Bayesian inference.

Discrete Distributions

Bernoulli Bern(p)

Isolated dual experiment: Success (probability p) and Failure $(1 - p)$.

Binomial Bin(n, p)



Accumulates the total count of independent successes under n identical iterations of Bernoulli variables. If p is tiny, it approximates Poisson; if n is massive, it approximates Gaussian.

Geometric Geom(p) / Negative Binomial NBin(r, p)

Register sequential Bernoulli iterations. The Geometric dictates failures prior to obtaining 1 success. The Negative Binomial expands the model by counting the failures required before registering r successes in the iteration.

Hypergeometric HGeom(w, b, n)

Sampling without replacement. Extracts n random objects from a finite universe of w desired specimens and b undesired ones. Lacks iterative independence.

Poisson Pois(λ)

Groups continuous flows over discrete domains. Typically used for extremely atypical events operating at temporally large scales or volumes with a strict expected rate.

Multivariate Distributions

Multinomial Mult$_k(n, \vec{p})$

Generalization of Binomial to vectors of k independent categories. Marginally they collapse back to the Binomial version. Joint PMF:

$$P(\vec{X} = \vec{n}) = \frac{n!}{\prod_{i=1}^k n_i!} \prod_{i=1}^k p_i^{n_i}$$

Multivariate Normal

Metric extension of Gaussian behavior in k-algebraic dimensions. Possesses the key property that uncorrelated vectors firmly equate to their underlying scalar independence. Its density function with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ is:

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^k |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

Conditional Bivariate Normal: For $(X_1, X_2) \sim \mathcal{N}$, the conditional expected value of X_1 given $X_2 = x_2$ is:

$$E(X_1 | X_2 = x_2) = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$$

$$\text{Var}(X_1 | X_2 = x_2) = \sigma_1^2 (1 - \rho^2)$$

Dirichlet Dir$(\vec{\alpha})$

Conjugate prior of the Multinomial. Models probability distributions over the standard simplex. Density proportional to $\prod_{i=1}^k x_i^{\alpha_i - 1}$.

Essential Properties

Elementary Linear Convolutions

Maintained under the axiom of independent variables X and Y:

1. $X \sim \text{Pois}(\lambda_1), Y \sim \text{Pois}(\lambda_2) \implies X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$
2. $X \sim \text{Bin}(n_1, p), Y \sim \text{Bin}(n_2, p) \implies X + Y \sim \text{Bin}(n_1 + n_2, p)$
3. $X \sim \text{Gamma}(a_1, \lambda), Y \sim \text{Gamma}(a_2, \lambda) \implies X + Y \sim \text{Gamma}(a_1 + a_2, \lambda)$
4. $X \sim \mathcal{N}(\mu_1, \sigma_1^2), Y \sim \mathcal{N}(\mu_2, \sigma_2^2) \implies X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
5. **Bank's Theorem (Beta-Gamma Relation):** $X \sim \text{Gamma}(a, \lambda), Y \sim \text{Gamma}(b, \lambda) \implies \frac{X}{X+Y} \sim \text{Beta}(a, b)$, where $(X + Y) \perp\!\!\!\perp \frac{X}{X+Y}$.

Analytical Inequalities

1. **Cauchy-Schwarz** $|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}$
2. **Markov** $P(|X| \geq a) \leq \frac{E|X|}{a}$
3. **Chebyshev** $P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$
4. **Jensen** $E(g(X)) \geq g(E(X))$ with convex g.

Entropy and Information

Shannon Entropy $H(X) = -\sum_x p(x)\log_2 p(x)$. Quantifies average uncertainty.

KL Divergence $D_{KL}(P||Q) = \sum_x P(x)\log_2\left(\frac{P(x)}{Q(x)}\right)$. Measures information loss when approximating P using Q.

Mutual Information $I(X;Y) = H(X) - H(X|Y)$

Cross Entropy $H(P,Q) = -\sum P(x)\log Q(x)$

Basic Formulas

Linear and Taylor Series

Geometric (Finite and Infinite): $\sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r}$ and asymptotic infinity for $|r| < 1$ as $\frac{1}{1-r}$.
Natural Exponential: $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$

Special Functions

Gamma Function: $\Gamma(t) = \int_0^\infty x^{t-1}e^{-x}dx$. Properties:
$\Gamma(t+1) = t\Gamma(t)$, $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$ and $\Gamma(1/2) = \sqrt{\pi}$.
Beta Function: $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.

Stirling's Approximation

Stirling's Formula: $n! \approx \sqrt{2\pi n}(n/e)^n$.

Exponential Family

A distribution belongs to the exponential family if its density can be written as:

$$f(x|\theta) = h(x)\exp\left(\eta(\theta)\cdot T(x) - A(\theta)\right)$$

Where $T(x)$ is the sufficient statistic, $\eta(\theta)$ the natural parameter and $A(\theta)$ the log-partition function.

Probability Generating Function (PGF)

For non-negative discrete R.V.s X:

$$G_X(z) = E(z^X) = \sum_{k=0}^{\infty} p_k z^k$$

Factorial moments: $E[X(X-1)\dots(X-k+1)] = G_X^{(k)}(1)$.

Statistical R Commands

Prefix	Action
`d...(x)`	Density or Mass (PDF/PMF)
`p...(q)`	Area under the curve (CDF)
`q...(p)`	Quantiles and probability cuts
`r...(n)`	Pseudo-Random Generator (i.i.d)

Replace ... with norm, binom, pois, exp, etc.

Supplementary Content

Common Mistakes and Supporting Notes

P-hacking Adjusting α given multiple tests (Bonferroni Correction: $\alpha_{adj} = \alpha/n$).

CI Interpretation Frequentist: The interval is random and the parameter fixed. Bayesian: The parameter is random (Credible Interval).

Expected Value Fallacy $E(g(X)) \neq g(E(X))$ unless g is linear (Jensen determines the direction of the inequality).

Independence vs. Correlation $\text{Corr}(X,Y) = 0 \not\Rightarrow X \perp\!\!\!\perp Y$. Uncorrelation only measures linear relationship.

Conditioning $P(A|B) \neq P(B|A)$ (Prosecutor's fallacy).

R.V. Support Check if the support depends on parameters (e.g. Unif$(0,\theta)$); this invalidates regularity conditions.

Cauchy: Possesses neither defined mean nor variance. Ratio of two independent standard Normals (Z_1/Z_2).

Memorylessness: Unique to Expo (cont.) and Geom (disc.).

Genesis of Distributions (Structural)

- χ_n^2: Sum of n squared standard Gaussians $\sum_{i=1}^n Z_i^2$.
- t_n: Ratio $Z/\sqrt{V/n}$ with $V \sim \chi_n^2$.
- F_{d_1,d_2}: Ratio of two normalized Chi-squares $(U/d_1)/(V/d_2)$.

MCMC Sampling (Metropolis-Hastings)

Acceptance rule to generate samples from a complex $P(x)$:

$$\alpha(x \to x') = \min\left(1, \frac{P(x')Q(x|x')}{P(x)Q(x'|x)}\right)$$

Asymptotic Approximations

Origin	Limit	Condition
Bin(n,p)	Pois$(\lambda = np)$	$n \to \infty, p \to 0$
Bin(n,p)	$\mathcal{N}(np, npq)$	$np > 5, nq > 5$
Pois(λ)	$\mathcal{N}(\lambda,\lambda)$	$\lambda > 20$
HGeom	Bin$(n, \frac{w}{w+b})$	$w + b \gg n$
t_n	$\mathcal{N}(0,1)$	$n \to \infty$

p-value Evidence (Fisher's Criterion)

p-value	Evidence against H_0
$p > 0.10$	No significant evidence
$0.05 < p \leq 0.10$	Marginal evidence (weak)
$0.01 < p \leq 0.05$	Moderate evidence
$p \leq 0.01$	Very strong evidence

Suggested Invariance and Inference Additions

Stochastic Decision Guide:
σ known $\implies Z$ (Normal).
σ unknown $(n < 30) \implies t$ (Student).
Var. Comparison $\implies F$ (Fisher).
Indep./Fit $\implies \chi^2$ (Chi-squared).

Useful Relations:
Bin$(1,p) \equiv$ Bern(p).
Gamma$(1,\lambda) \equiv$ Expo(λ).
Gamma$(n/2, 1/2) \equiv \chi_n^2$.

Variance of Subtraction:
$\text{Var}(aX - bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) - 2ab\text{Cov}(X,Y)$.

FPCF (Sampling w/o repl.): Finite population correction factor: $\sqrt{\frac{N-n}{N-1}}$.

Beta Bayes Interpretation: In Beta(α,β), $\alpha - 1$ are prior successes and $\beta - 1$ are prior failures.

Useful Matrix Identities

In the context of the Multivariate Normal $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$:

1. $(\mathbf{X}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) \sim \chi_k^2$.
2. $\text{Var}(\mathbf{AX}+\mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$.
3. If $\boldsymbol{\Sigma}$ is diagonal $\implies$ independent components.

Inference and Advanced Methods

Cramér-Rao Bound: $\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$ for unbiased estimators.

Wilks' Theorem: Under H_0, the likelihood ratio statistic $\lambda \xrightarrow{D} \chi_d^2$.
Bayes Kernel: Posterior $\propto$ Likelihood $\times$ Prior.
Multi Delta Method: If $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{D} \mathcal{N}(0,\Sigma)$:

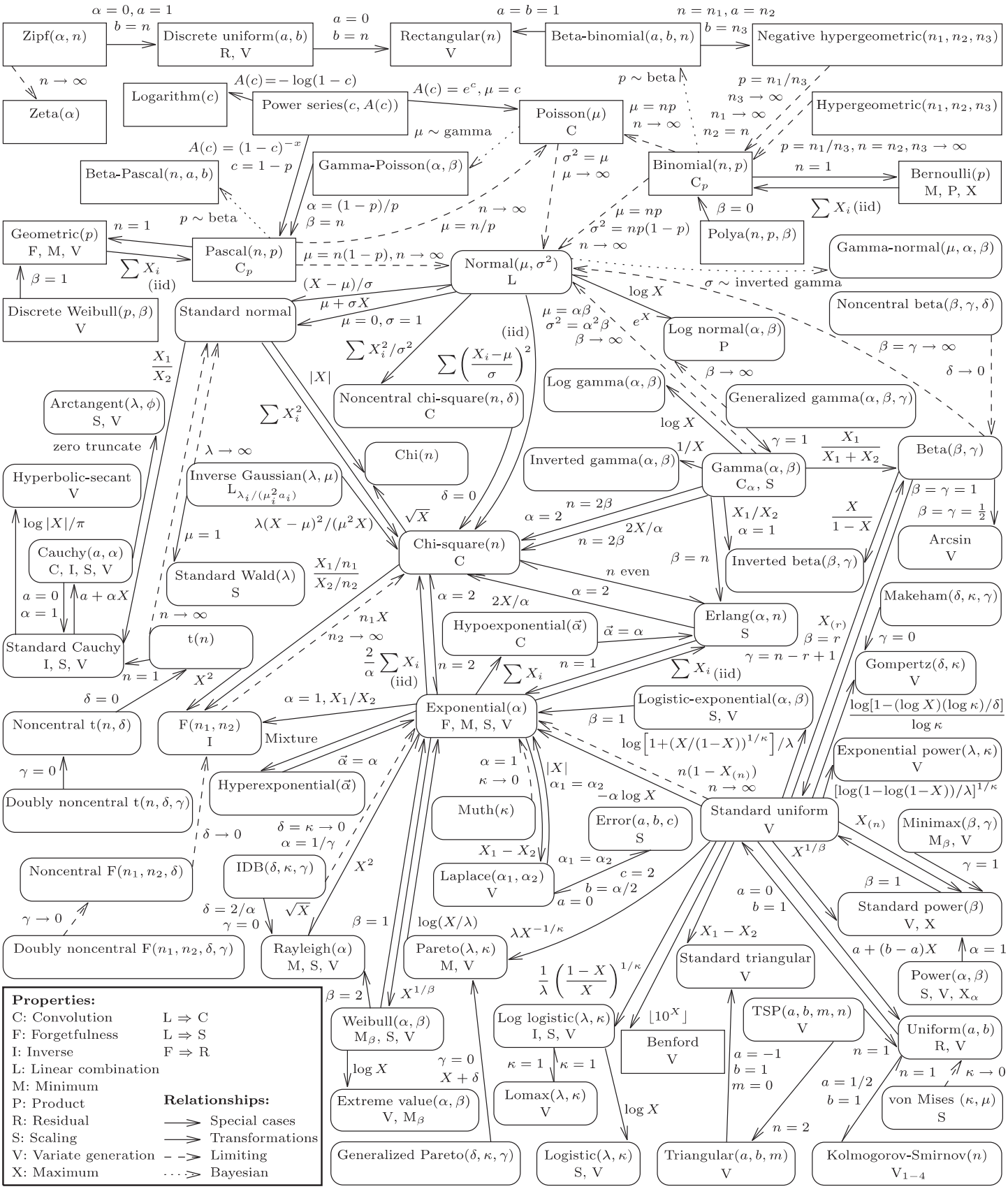
$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{D} \mathcal{N}(0, \nabla g(\theta)^T\Sigma\nabla g(\theta))$$

Convergence Hierarchy

$$\begin{tikzcd} & a.s. \arrow[d] & \\ L_2 \arrow[r] & P \arrow[r] & D \end{tikzcd}$$

Table of Distributions

Distribution	PMF/PDF and Domain (Support)	Expectation	Variance	MGF
Bernoulli $Bern(p)$	$P(X=1)=p, P(X=0)=q=1-p$	p	pq	$q+pe^t$
Binomial $Bin(n,p)$	$P(X=k)=\binom{n}{k}p^k q^{n-k}$ $k \in \{0,1,\dots,n\}$	np	npq	$(q+pe^t)^n$
Geometric $Geom(p)$	$P(X=k)=q^k p$ (failures) $k \in \{0,1,\dots\}$	q/p	q/p^2	$\frac{p}{1-qe^t}$
Negative Binomial $NBin(r,p)$	$P(X=n)=\binom{n+r-1}{n}p^r q^n$ $n \in \{0,1,\dots\}$	rq/p	rq/p^2	$(\frac{p}{1-qe^t})^r$
Hypergeometric $HGeom(w,b,n)$	$P(X=k)=\frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}$ $k \in \{0,1,\dots,n\}$	$\frac{nw}{w+b}$	$n\frac{w}{w+b}\frac{b}{w+b}\frac{w+b-n}{w+b-1}$	Intractable
Poisson $Pois(\lambda)$	$P(X=k)=\frac{e^{-\lambda}\lambda^k}{k!}$ $k \in \{0,1,\dots\}$	λ	λ	$e^{\lambda(e^t-1)}$
Uniform $Unif(a,b)$	$f(x)=\frac{1}{b-a}$ $x \in (a,b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Normal $N(\mu,\sigma^2)$	$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$ $x \in (-\infty,\infty)$	μ	σ^2	$e^{\mu t+\frac{\sigma^2 t^2}{2}}$
Exponential $Expo(\lambda)$	$f(x)=\lambda e^{-\lambda x}$ $x \in [0,\infty)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Gamma $Gamma(\alpha,\lambda)$	$f(x)=\frac{\lambda^\alpha x^{\alpha-1}e^{-\lambda x}}{\Gamma(\alpha)}$ $x \in [0,\infty)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$(\frac{\lambda}{\lambda-t})^\alpha$
Beta $Beta(a,b)$	$f(x)=\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}x^{a-1}(1-x)^{b-1}$ $x \in (0,1)$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2(a+b+1)}$	Intractable
Log Normal $LN(\mu,\sigma^2)$	$f(x)=\frac{1}{x\sigma\sqrt{2\pi}}e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}$ $x>0$	$e^{\mu+\sigma^2/2}$	$(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$	Does not exist
Weibull $W(k,\lambda)$	$f(x)=\frac{k}{\lambda}(\frac{x}{\lambda})^{k-1}e^{-(x/\lambda)^k}$ $x \geq 0$	$\lambda\Gamma(1+1/k)$	$\lambda^2[\Gamma(1+2/k)-\Gamma^2(1+1/k)]$	See Stirling
Pareto $Pa(x_m,\alpha)$	$f(x)=\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$ $x \geq x_m$	$\frac{\alpha x_m}{\alpha-1}(\alpha>1)$	$\frac{x_m^2\alpha}{(\alpha-1)^2(\alpha-2)}(\alpha>2)$	Does not exist
Laplace $L(\mu,b)$	$f(x)=\frac{1}{2b}e^{-\frac{\lvert x-\mu\rvert}{b}}$ $x \in (-\infty,\infty)$	μ	$2b^2$	$\frac{e^{\mu t}}{1-b^2t^2}(\lvert t\rvert<1/b)$
Chi-Squared χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$ $x \in (0,\infty)$	n	$2n$	$(1-2t)^{-n/2}$
Student's t t_ν	$f(x) \propto (1+\frac{x^2}{\nu})^{-\frac{\nu+1}{2}}$	$0(\nu>1)$	$\frac{\nu}{\nu-2}(\nu>2)$	Intractable
Fisher's F F_{d_1,d_2}	$f(x) \propto \frac{\sqrt{\frac{(d_1x)^{d_1}d_2^{d_2}}{(d_1x+d_2)^{d_1+d_2}}}}{xB(\frac{d_1}{2},\frac{d_2}{2})}$	$\frac{d_2}{d_2-2}$	$\frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}$	Intractable
Cauchy $Cauchy(x_0,\gamma)$	$f(x)=\frac{1}{\pi\gamma[1+(\frac{x-x_0}{\gamma})^2]}$ $x \in (-\infty,\infty)$	Undefined	Undefined	Undefined


Zipf(α, n)
Discrete uniform(a, b) R, V
Rectangular(n) V
Beta-binomial(a, b, n)
Negative hypergeometric(n1, n2, n3)
Zeta(α)
Logarithm(c)
Power series(c, A(c))
Poisson(μ) C
Hypergeometric(n1, n2, n3)
Beta-Pascal(n, a, b)
Gamma-Poisson(α, β)
Binomial(n, p) Cp
Bernoulli(p) M, P, X
Geometric(p) F, M, V
Pascal(n, p) Cp
Polya(n, p, β)
Gamma-normal(μ, α, β)
Normal(μ, σ²) L
Discrete Weibull(p, β) V
Standard normal
Noncentral beta(β, γ, δ)
Log normal(α, β) P
Log gamma(α, β)
Generalized gamma(α, β, γ)
Arctangent(λ, φ) S, V
Noncentral chi-square(n, δ) C
Chi(n)
Inverted gamma(α, β)
Gamma(α, β) Cα, S
Beta(β, γ)
Hyperbolic-secant V
Inverse Gaussian(λ, μ) Lλi/(μi²ai)
Cauchy(a, α) C, I, S, V
Standard Wald(λ) S
Chi-square(n) C
Inverted beta(β, γ)
Arcsin V
Makeham(δ, κ, γ)
Standard Cauchy I, S, V
t(n)
Hypoexponential(α⃗) C
Erlang(α, n) S
Gompertz(δ, κ) V
Noncentral t(n, δ)
F(n1, n2) I
Exponential(α) F, M, S, V
Logistic-exponential(α, β) S, V
Exponential power(λ, κ) V
Doubly noncentral t(n, δ, γ)
Hyperexponential(α⃗)
Muth(κ)
Error(a, b, c) S
Standard uniform V
Minimax(β, γ) Mβ, V
Noncentral F(n1, n2, δ)
IDB(δ, κ, γ)
Laplace(α1, α2) V
Standard power(β) V, X
Doubly noncentral F(n1, n2, δ, γ)
Rayleigh(α) M, S, V
Pareto(λ, κ) M, V
Standard triangular V
Power(α, β) S, V, Xα
Weibull(α, β) Mβ, S, V
Log logistic(λ, κ) I, S, V
Benford V
TSP(a, b, m, n) V
Uniform(a, b) R, V
Extreme value(α, β) V, Mβ
Lomax(λ, κ) V
von Mises (κ, μ) S
Generalized Pareto(δ, κ, γ)
Logistic(λ, κ) S, V
Triangular(a, b, m) V
Kolmogorov-Smirnov(n) V1−4
Properties:
C: Convolution
F: Forgetfulness
I: Inverse
L: Linear combination
M: Minimum
P: Product
R: Residual
S: Scaling
V: Variate generation
X: Maximum
L ⇒ C
L ⇒ S
F ⇒ R
Relationships:
Special cases
Transformations
Limiting
Bayesian